Maximum Blackouts That Keep Every Lattice Rectangle Identifiable

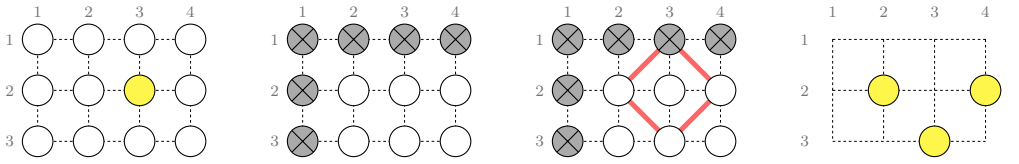
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Abstract

Kagey's Problem 001 concerns a grid of points, some of them blacked out. A rectangle with its corners on grid points, tilted or not, shows only its white corners. We ask how many points can be blacked out while every rectangle stays determined by what it shows. We study the maximum blackout and the number of optima, tabulated in A397315 and A399580 of the On-Line Encyclopedia of Integer Sequences. A row-column graph and Mantel's theorem bound the maximum by the width plus a quarter of the squared height. We find the maximum on up to five rows at every width, and on six rows from width 15 on. The five-row result refutes our earlier guess that the maximum is the sum of the two sides. We count the three-row optima exactly and find the growth rate on four rows. At every fixed height, the maximum minus the width is eventually periodic, and the optimum counts have a rational generating function. Some values rest on computer checks.

1 Introduction

You black out some points of the $h \times m$ grid, and we call this set the *blackout*. A game master draws a rectangle, tilted or not, with corners on grid points,



(a) the point $(3, 2)$ (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on a largest valid blackout of the 3×4 grid (Section 5). Points are (column, row). Gray crossed circles are black. Panel (c) shows the red rectangle, and (d) its yellow corners. The three yellow corners fix the fourth by Lemma 4.

and says which corners are white. You must name all four. Kagey [16, 17] asks for the largest *valid* blackout, where no two rectangles look the same (Definition 1). Figure 1 plays one round.

Let $f(h, m)$ be the largest size of a valid blackout, and $a_h(m)$ the number of these. Our OEIS entries A397315 and A399580 [29] first recorded the conjecture $f(3, m) = m + 2$, which the Omega Institute [25] proved in Lean, and the guess $f(h, m) = h + m$ for $\min(h, m) \geq 4$ apart from $f(4, 4) = 7$ and $f(4, 5) = 8$. The guess first fails at 5×8 (Theorem 39). Put $q_h = \lfloor h^2/4 \rfloor$. Results that rest on a computer check are marked computer-assisted in their statements.

Theorem A (narrow grids)**.**

(a) We have $f(h, m) \geq h + m - 1$, and $f(h, m) \leq m + q_h$ if $m \geq 3$.

(b) We have $f(2, m) = m + 1$ and $f(3, m) = m + 2$ for $m \geq 3$, $f(4, m) = m + 4$ for $m \geq 6$, and $f(5, 5) = 10$.

(c) (Computer-assisted, with checked certificates for $f(5, 6)$ and $f(5, 7)$, and the last claim checked in Lean.) We have $f(4, 4) = 7$, $f(4, 5) = 8$, $f(5, 6) = 11$, $f(5, 7) = 12$, and $f(5, m) = m + 6$ for $m \geq 8$.

(d) (Computer-assisted for $m \geq 9$, with checked certificates for $m \leq 14$ and checked in Lean for $m \geq 15$.) We have $f(6, m) \leq m + 8$ for $8 \leq m \leq 14$, and $f(6, m) = m + 9$ for $m \geq 15$.

(e) (Computer-assisted, with checked certificates.) We have $f(6, 6) = 12$, $f(6, 7) = 13$, $f(7, 7) = 14$, and $f(6, 8) = 15$.

Part (a) is Lemma 16 and Theorem 8, (b) is Theorem 5, Section 5, and Theorems 31 and 43, (c) is Corollary 32 and Theorem 39, (d) is Theorem 41, and (e) is Theorem 42. For (c) and (d) the finite collar theorem (Theorem 38) reduces validity at every larger width to one finite board.

Theorem B (three-row optima)**.** *For $m \geq 4$ we have $a_3(m) = 4(m-1)(3m+2)\,3^{m-5} + E_m$, where $E_m > 0$ counts the optima with a column black in rows 1 and 3, and $E_m = O(m^2(1+\sqrt{3})^m)$. So $a_3(m) \sim \frac{4}{81} m^2 3^m$, and a fraction tending to $4/27$ of the $m^2 3^{m-1}$ axis-valid candidates is valid. The series $\sum_{m\geq 3} a_3(m) z^m$ is rational. (Computer-assisted.) Its reduced denominator is $(1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$, and $E_m \sim \frac{19+11\sqrt{3}}{432} m^2(1+\sqrt{3})^m$.*

See Theorems 23, 24, and 25, and Section 5.2.

Theorem C (four rows)**.** *If $m \geq 4$, no valid blackout of the $4 \times m$ grid hides all corners of a rectangle. The series $\sum_{m\geq 6} a_4(m) z^m$ is rational with reduced denominator of degree 420, and $a_4(m) \sim c\, m^4 \rho_4^m$ with $c \approx 8.1611 \times 10^{-5}$ and $\rho_4 = 3.35199\ldots$, an algebraic number of degree 20. (Computer-assisted.)*

See Theorems 33 and 36 and Corollary 37.

Theorem D (every fixed height)**.** *Fix $h \geq 1$ and write a blackout as the word $B_1 \cdots B_m$ of the black rows of its columns. The valid words form a regular language. If $h \geq 2$, then $f(h,m) - m$ is eventually periodic in m, and $\sum_{m\geq 0} a_h(m) z^m$ is rational.*

See Theorems 44 and 45.

Prior work. The game asks for a test cover [3, 11, 18, 23] (Corollary 3). Forbidding black axis-aligned rectangles is the Zarankiewicz problem [28]. Distinguishing them forbids four- and six-cycles in the row-column graph (Lemma 7), and de Caen and Székely [6, 7] study such graphs with unbalanced parts. Neuwirth [24, Theorem 1(ii) and Proposition 17] bounds their edges, so the row bound is known in graph form. Our proof with Mantel's theorem [19] also finds and counts the equality case. None of these works sees tilted rectangles. Column transfer [1, 8, 20], [9, Proposition V.6] handles fixed heights, and Theorem 1(3) of Guilhot, Little, and Parkinson [12] with Theorem 44 also gives Theorem D.

Outline. Sections 2 to 4 prove the general bounds, Sections 5 to 7 treat three to six rows with the finite collar theorem in Section 7, and Section 8 treats every fixed height. Section 9 describes the computations, Section 10 lists open questions, and Appendix A treats square grids.

2 Conventions

(1) An $h \times m$ grid has h rows, m columns, and point set V. A $2 \times m$ grid is a *strip*. The point (i, j) lies in column i and row j, with $(1, 1)$ at the top left (Figure 1(a)).

(2) A *rectangle* R is its set $C(R)$ of four corners, grid points spanning a rectangle of positive area. It is *axis-aligned* if its sides are parallel to the grid lines, and *tilted* otherwise.

(3) A *blackout* is a set $S \subseteq V$ with *kept set* $K = V \setminus S$. The *presentation* of R is $P_S(R) = C(R) \setminus S$.

(4) Column i has *black rows* $B_i = \{j : (i, j) \in S\}$, and is *empty*, a *singleton*, a *double*, or a *triple* when $|B_i|$ is 0, 1, 2, or 3, and *full* when $|B_i| = h$. The *pair set* F is the set of row pairs contained in some B_i, and $q_h = \lfloor h^2/4 \rfloor$ is A002620(h).

Definition 1 (valid blackout)**.** A blackout S is *valid* if $P_S(R_1) \neq P_S(R_2)$ whenever $R_1 \neq R_2$. The *maximum blackout* $f(h, m)$ is the largest size of a valid blackout of the $h \times m$ grid, and $a_h(m)$ counts the valid blackouts of that size, with their coordinates.

By symmetry $f(h, m) = f(m, h)$ and $a_h(m) = a_m(h)$. Width 0 and height 1 have no rectangles, so $f(h, 0) = 0$, $a_h(0) = 1$, $f(1, m) = m$, and $a_1(m) = 1$. An empty presentation is allowed, so a valid blackout hides at most one rectangle.

Lemma 2 (difference sets)**.** *For sets A, B, and S, we have $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$.*

Proof. Both conditions say that no point outside S lies in exactly one of A and B. □

Corollary 3 (hitting sets)**.** *A blackout S is valid if and only if K meets $C(R_1) \triangle C(R_2)$ for all distinct rectangles R_1 and R_2.*

Proof. Apply Lemma 2 to $C(R_1)$ and $C(R_2)$. □

So subsets of valid blackouts are valid, and showing that no blackout of k points is valid proves $f(h, m) < k$.

Lemma 4 (three corners suffice)**.** *Three corners of a rectangle determine it, so distinct rectangles with the same presentation show at most two corners.*

Proof. Three corners form a right triangle, whose only right angle is at the corner q adjacent to the other two, p and r. So the fourth corner is $p + r - q$. □

3 Strips

On two rows every rectangle is axis-aligned, since a tilted one meets three rows: the neighbors of a lowest corner lie strictly above it, as no side is horizontal, and the opposite corner lies above them.

Theorem 5 (strip theorem)**.** *Let $m \geq 3$.*

(a) A blackout of the $2 \times m$ strip is valid if and only if at most one of its columns is full.

(b) We have $f(2, m) = m + 1$, and the maximum blackouts are those with one full column and one black point in every other column.

(c) There are $a_2(m) = m \cdot 2^{m-1}$ maximum blackouts, so row 2 of A399580 is A001787 from $m = 3$ on.

For $m = 2$ there is one rectangle, so $f(2, 2) = 4$ and $a_2(2) = 1$.

Proof. (a) Distinct rectangles share at most one column, so their difference set contains all four points of two columns p and q. Every pair arises, since $m \geq 3$ gives a third column z and the rectangles on $\{z, p\}$ and $\{z, q\}$. By Corollary 3, S is valid if and only if K meets every two columns, that is, at most one column is full.

(b, c) By (a), $|K| \geq m - 1$, so $|S| \leq m + 1$, with equality exactly for the blackouts in (b). There are m choices of full column and two in each other column. □

Proposition 6 (orbits and kings)**.** *For $m \geq 3$ the number o_m of maximum blackouts of the $2 \times m$ strip up to the row reflection and the column reversal is $m\,2^{m-3} + [m \text{ odd}]\,2^{(m-3)/2}$. It equals* A322284*$(m-1)$, which counts $m-1$ nonattacking kings on a $2 \times (2m-2)$ board up to symmetry.*

Proof. By Burnside's lemma, $4o_m$ is $a_2(m)$ plus the optima fixed by the reflection (none) and by the reversal and the half turn (each fixes none for even m, and $2^{(m-1)/2}$ with a central full column for odd m).

Put $n = m-1$. Nonattacking kings on the $2 \times 2n$ board are columns $c_1 < \cdots < c_n$ with $c_{j+1} \geq c_j + 2$, and any rows r_j. The gaps $c_1 - 1$, $c_{j+1} - c_j - 2$, and $2n - c_n$ are nonnegative and sum to 1, so exactly one, say the eth, is 1, and $c_j = 2j - 1 + [j \geq e]$. Sending this to the optimum with full column e and other black points in rows $r_1, \ldots, r_n$ from left to right is a bijection. It commutes with the reflection, which swaps the rows on both sides, and with the reversal, which sends e to $n + 2 - e$ and c to $2n + 1 - c$ and reverses the rows, since $2n + 1 - c_{n+1-j} = 2j - 1 + [j \geq n + 2 - e]$. So the orbits correspond. □

4 The row-column graph and general bounds

A blackout S is *axis-valid* if no two axis-aligned rectangles have the same presentation. A rectangle showing at most two corners is *weak*. The *row-column graph* G_S has vertices r_j for the rows and c_i for the columns, and an edge $r_j c_i$ for each black point (i, j).

Lemma 7 (axis-aligned rectangles and short cycles)**.** *If G_S has no four-cycle and no six-cycle, then S is axis-valid. The converse holds for $m \geq 3$.*

Proof. For the converse, let $m \geq 3$. A four-cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ and a third column w make $\{i, w\} \times \{j, j'\}$ and $\{i', w\} \times \{j, j'\}$ show the same points. A six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ makes $\{x, y\} \times \{a, b\}$ and $\{y, z\} \times \{a, c\}$ black except possibly at (y, a).

For the first statement, let $R_1 \neq R_2$ be axis-aligned with the same presentation, so $C(R_1) \triangle C(R_2) \subseteq S$ (Lemma 2). Opposite corners determine a rectangle, so by symmetry the common corners lie in one row a. With none, R_1 is a black four-cycle. Two, (x, a) and (y, a), leave the other rows $b \neq c$ and the black four-cycle $r_b\, c_x\, r_c\, c_y$. One, (x, a), with $R_1 = \{x, y\} \times \{a, b\}$ and $R_2 = \{x, z\} \times \{a, c\}$, gives the black six-cycle $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$. □

Theorem 8 (row bound)**.** *Let $m \geq 3$. Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points, and at most $h + q_m$ if also $h \geq 3$. Hence $f(h, m) \leq m + q_h$.*

For each nonempty column i choose a tree T_i on B_i, and let H be their union.

Lemma 9 (row graph)**.** *Let $m \geq 3$ and let S be axis-valid. For any choice of the trees, H is simple and triangle-free, and $e(H) = |S| - t$, where t is the number of nonempty columns.*

Proof. By Lemma 7, G_S has no four-cycle, so H has no repeated edge and $e(H) = \sum(|B_i| - 1) = |S| - t$. A triangle $r_a r_b r_c$ is not in one tree. Edges from two columns i, i' put two of a, b, c in $B_i \cap B_{i'}$, and edges from three columns give a six-cycle $r_a\, c_i\, r_b\, c_{i'}\, r_c\, c_{i''}$. □

Lemma 10 (Mantel [19])**.** *Let H be a simple triangle-free graph on h vertices, let a vertex of maximum degree Δ have neighbor set A, and let B be the other $h - \Delta$ vertices, spanning $e(B)$ edges. Then*

$$e(H) = \sum_{b \in B} d(b) - e(B) \leq (h - \Delta)\Delta - e(B) \leq q_h.$$

Proof. Every edge meets B, as A is independent, and the degree sum over B counts the edges inside B twice. Also $d(b) \leq \Delta$ and $(h - \Delta)\Delta \leq q_h$. □

Proof of Theorem 8. Lemmas 9 and 10 give $|S| = t + e(H) \leq m + q_h$. Only Lemma 7 uses a third column, so for $h \geq 3$ the transpose gives $|S| \leq h + q_m$. □

Theorem 11 (equality structure)**.** *Let $h \geq 4$ and $m \geq 3$. A blackout of the $h \times m$ grid with $m + q_h$ points is axis-valid if and only if*

(a) every column is a singleton or a double,

(b) the q_h doubles have distinct row pairs, and

(c) these pairs form a balanced complete bipartite graph on the rows.

There are $p_h\,(m)_{q_h}\,h^{m-q_h}$ such blackouts, where $(m)_q$ is the falling factorial and the number p_h of balanced bipartitions of the rows is $\binom{h}{\lfloor h/2 \rfloor}$, halved for even h.

Proof. Let S be axis-valid. Equality in Lemmas 9 and 10 forces $t = m$, $e(B) = 0$, and $(h - \Delta)\Delta = q_h$ with B joined to all of A, so H is as in (c). If $|B_i| \geq 3$, some v has two neighbors u, w in T_i, and as $h \geq 4$ the part of v has another vertex z. Replacing vw by uw in T_i keeps uz and wz and makes the triangle uwz, against Lemma 9. This gives (a), and the edges of H, the row pairs of the doubles, give (b) and (c).

Conversely, (a) and (b) rule out four-cycles, and a six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ would need the doubles $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$, a triangle in (c), so Lemma 7 applies. The blackout determines, and is determined by, a bipartition, an injective map from its edges to columns, and a row for each other column. □

Corollary 12 (small widths)**.** *Let $h \geq 4$ and $3 \leq m < q_h$. Then $f(h, m) \leq m + q_h - 1$. In particular $f(5, 5) \leq 10$.*

Proof. By Theorems 8 and 11, a valid blackout of $m + q_h$ points has $q_h \leq m$ doubles. Here $q_5 = 6$. □

Put $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$, so $W_3 = 2$, $W_4 = 3$, and $W_5 = 5$.

Lemma 13 (leftmost corner)**.** *A tilted rectangle R has a unique corner $\ell = (x, r)$ in its smallest column, and a unique corner in its smallest row. The sides from ℓ are $(a, -b)$ and (c, d) with positive integers a, b, c, d and $ac = bd$, and the corners are*

$$(x, r), \qquad (x + a,\, r - b), \qquad (x + c,\, r + d), \qquad (x + a + c,\, r - b + d). \quad (1)$$

Its rows are $r - b$ to $r + d$, and its span *is $a + c$.*

Proof. No side is vertical, so two corners in column x would be opposite, and the other two, with mean column x, would lie in column x too. Rows are the same. So both sides from ℓ go right, one up and one down, and perpendicularity gives $ac = bd$. □

Lemma 14 (span bound)**.** *Every tilted rectangle of the $h \times m$ grid has span $a + c \leq W_h$.*

Proof. Here $b + d \leq h - 1$, and $(a - 1)(c - 1) \geq 0$ gives $a + c \leq bd + 1 \leq W_h$, since bd is an integer at most $(b + d)^2/4$. □

So W_h is A033638$(h - 1)$, attained for $h \geq 3$ and $m > W_h$ by the sides $(1, -b)$ and (bd, d) with $b + d = h - 1$ and $|b - d| \leq 1$. The same argument proves the conjecture in A338885 that its row n has largest entry A033638(n).

The *diamond* $D(i,j)$ is the square with corners $(i,j+1)$, $(i+1,j)$, $(i+2,j+1)$, and $(i+1,j+2)$. The rectangles A_i, B_i, C_i, and E_i have one corner in each of the columns i to $i+3$, in the rows 3412, 3142, 2413, and 2143 from left to right.

Lemma 15 (tilted rectangles on narrow grids)**.** *The tilted rectangles of the $3 \times m$ grid are the diamonds $D(i,1)$. Those of the $4 \times m$ grid are $D(i,1)$, $D(i,2)$, A_i, B_i, C_i, and E_i.*

Proof. Each listed set is a rectangle. On three rows $b = d = 1$, so $a = c = 1$ and ℓ is in row 2. On four rows either $b = d = 1$, which gives the diamonds, or $\{b,d\} = \{1,2\} = \{a,c\}$ and ℓ is in row $1+b$. The sides $(2,-2),(1,1)$; $(1,-2),(2,1)$; $(2,-1),(1,2)$; and $(1,-1),(2,2)$ give A, B, C, and E. □

Lemma 16 (boundary cross)**.** *The first row and the first column of the $h \times m$ grid form a valid blackout, so $f(h,m) \geq h + m - 1$.*

Proof. By Lemma 4 only presentations of at most two points matter. An axis-aligned rectangle $\{x_1,x_2\} \times \{y_1,y_2\}$ shows $X \times Y$ with $X = \{x_1,x_2\} \setminus \{1\}$ and $Y = \{y_1,y_2\} \setminus \{1\}$, which determine it, since $|X| = 1$ exactly when $1 \in \{x_1,x_2\}$, and likewise for Y. By Lemma 13 a tilted rectangle has at most two black corners, ℓ and $(x+a, r-b)$. If both are black, then $\ell = (1, 1+b)$ and it shows $p = (1+c, 1+b+d)$ and $q = (1+a+c, 1+d)$. These determine a,b,c,d and differ in both coordinates, so no axis-aligned rectangle shows $\{p,q\}$. □

Lemma 17 (axis-aligned rivals)**.** *Let no axis-aligned rectangle be fully black. A* rival *of a set P of at most two white points is an axis-aligned rectangle that shows exactly P. The empty set has no rival, and P has one exactly when it is a singleton $\{(x,r)\}$ with $\{r,s\} \in F$ for some $s \in B_x$, a vertical pair $\{(x,r),(x,s)\}$ with $\{r,s\} \in F$, a horizontal pair $\{(x,r),(z,r)\}$ with $B_x \cap B_z \neq \emptyset$, or an oblique pair $\{(x,r),(z,s)\}$ with $x \neq z$, $r \neq s$, and $(x,s),(z,r) \in S$.*

Proof. A rival of $P \neq \emptyset$ has rows $\{r,s\}$, columns $\{x,z\}$, and its other corners black. As (x,r) is white, this is the listed condition, which forces $z \neq x$ and $s \neq r$. □

5 Three rows

Throughout, $m \geq 3$, the rows are $1, 2, 3$ from the top, and the tilted rectangles are the diamonds $D(i, 1)$ (Lemma 15). Theorem 8 and Lemma 16 give $f(3, m) = m + 2$. This is the result of [25]. The columns are $a = \{1\}$, $b = \{2\}$, $c = \{3\}$, $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$ and $X = \{1, 2, 3\}$, with a and c *outer*. A blackout without empty columns is its *word* $B_1 \cdots B_m$, and each pair of F (Section 2) is named by its double letter.

Proposition 18 (the two types)**.** *Let S be a blackout of the $3 \times m$ grid with $m + 2$ points.*

(a) If S is valid, its word is one X among singletons (type T) or two different doubles among singletons (type P).

(b) For every word of type T or P, G_S is a tree, so axis-aligned rectangles have distinct presentations.

(c) There are $m3^{m-1} + \binom{m}{2} \cdot 6 \cdot 3^{m-2} = m^2 3^{m-1}$ words of type T or P.

Proof. (a) By Lemmas 9 and 10, $|S| = t + e(H)$ with $t \leq m$ nonempty columns and $e(H) \leq 2$. So every column is nonempty and $\sum_i (|B_i| - 1) = 2$, and two equal doubles would form a four-cycle in G_S, against Lemma 7. (b) The column X, or the two doubles, connect the three rows, and every other column hangs on one row. (c) Place X, or two different doubles, and choose a singleton for each other column. □

These words are the *candidates*, in the class X and the classes AC, CA, AB, BA, BC and CB named by their doubles from left to right. Let AC_m hold the valid words of width m in AC, and likewise for the others.

Lemma 19 (diamond test)**.** *A candidate S with pair set F is valid if and only if every diamond $D(i, 1)$ passes the test below, where $P = D(i, 1) \setminus S$.*

(a) If $|P| \geq 3$ or P is empty, the diamond passes.

(b) If $P = \{(i + 1, 1), (i + 1, 3)\}$, the diamond fails if and only if $B \in F$.

(c) If $P = \{(i, 2), (i + 2, 2)\}$, it fails if and only if $B_i \cap B_{i+2}$ meets $\{1, 3\}$.

(d) If P is an oblique pair $\{(x, r), (z, s)\}$, it fails if and only if $(x, s), (z, r) \in S$.

(e) If $P = \{(x, r)\}$, it fails if and only if $\{r, s\} \in F$ for some $s \in B_x$.

Proof. Axis-aligned rectangles are told apart (Proposition 18(b)), three points fix a rectangle (Lemma 4), and an empty diamond has no axis-aligned rival, as G_S has no four-cycle. Tests (b) to (e) are Lemma 17, a rival needing $B \in F$ in (b) and $\{r, s\} \in F$ in (e). Two diamonds share at most a tip, so a common presentation would need both middle columns black in rows 1 and 3, which no candidate has. □

A *window* is three consecutive letters, at columns i to $i + 2$, and *passes* or *fails* with $D(i, 1)$. Its *tips* $(i, 2)$ and $(i + 2, 2)$ are black exactly when their letters are b, A, C or X.

Lemma 20 (singleton dichotomy)**.** *In a candidate, a window of three singletons fails if and only if $B \in F$ and it is bab, bbb or bcb.*

Proof. With both tips black, the diamond shows the vertical pair or one point of the middle column, and tests (b) and (e) fail it exactly when $B \in F$. With one black tip it shows three corners or an oblique pair passing test (d), as $(i + 1, 2)$ is white. With no black tip at least three corners show. □

Lemma 21 (six of nine in the class $\{A, C\}$)**.** *Let S be a candidate with $F = \{A, C\}$, and let s, t be singletons.*

(a) A window sAt fails if and only if st is bb, bc or cb, also for $t = C$ read as $t = b$.

(b) A window sCt fails if and only if st is bb, ba or ab.

(c) A window with one double, at a tip, passes.

Proof. (a) Column $i + 1$ shows only $(i + 1, 3)$, and the tip $(i + 2, 2)$ is black for $t \in \{b, C\}$. With both tips black, test (e) fails, as $2 \in B_{i+1}$ and $C \in F$. With only $(i+2, 2)$ black, the diamond shows $(i, 2)$ and $(i+1, 3)$, and test (d) fails it exactly when $s = c$. The case of $(i, 2)$ alone is symmetric, and with no black tip three corners show. Reflection in the middle row gives (b). (c) For Ast the proof of Lemma 20 applies with the tip $(i, 2)$ in place of $s = b$, as $B \notin F$. Reflection and reversal give the other windows. □

By Lemma 19, the set AC_m holds the words $UAVCW$ of width m, with U, V and W singleton strings, whose windows pass. Put

$G_{AC}(z) = \sum_{m\ge 2} |AC_m| z^m$, define G_{AB} likewise from the words $UAVBW$, and let $G_X(z) = \sum_{m\ge 1} |X_m| z^m$ come from the words UXW. With $R = 1 - 3z$, nonempty singleton strings with a fixed letter next to a double have generating function z/R.

Proposition 22 (the class $\{A, C\}$)**.** *The generating function of AC is*

$$G_{AC}(z) = \frac{z^2 - 4z^3 + 10z^4 - 8z^5 - 3z^6}{(1-3z)^3}, \tag{2}$$

and

$$|AC_m| = 2(m-1)(3m+2)3^{m-5} \qquad (m \ge 4). \tag{3}$$

Proof. By Lemma 21(a) and the half turn, a window tAC passes only for $t = a$ and ACt only for $t = c$, and AtC passes by tests (b) and (e). With Lemmas 20 and 21(c), only windows with a double in the middle can fail. If V is empty, a nonempty U ends in a and a nonempty W begins with c, giving $z^2(1+z/R)^2$. If $V = t$, Lemma 21(a),(b) leaves $p_t = 3, 1, 2$ last letters for U and $q_t = 2, 1, 3$ first letters for W, for $t = a, b, c$. As $\sum p_t = \sum q_t = 6$ and $\sum p_t q_t = 13$, this gives $z^3 \sum_t (1 + p_t z/R)(1 + q_t z/R) = z^3(3 + 12z/R + 13z^2/R^2)$. If $|V| \ge 2$, each end gives $3 + 6z/R$, for v_1 alone or one of the six passing pairs (u, v_1), and the middle $1/R$, for $9z^4(1-z)^2/R^3$. The three cases sum to (2), and $[z^m]z^k R^{-3} = \binom{m-k+2}{2} 3^{m-k}$ gives (3). □

Theorem 23 (the number of three-row optima)**.** *Let E_m count the optima of the $3 \times m$ grid with a column black in rows 1 and 3. For $m \ge 4$,*

$$a_3(m) = 4(m-1)(3m+2)3^{m-5} + E_m. \tag{4}$$

Proof. Only B and X are black in rows 1 and 3, so by Proposition 18(a) these optima are those outside AC and CA. Column reversal maps AC, AB and BC onto CA, BA and CB, and the reflection in the middle row maps AB onto CB, so

$$a_3(m) = 2|AC_m| + 4|AB_m| + |X_m| = 2|AC_m| + E_m \qquad (m \ge 3), \tag{5}$$

and (3) gives (4). □

5.1 The generating function

Put $L = 1 - 2z - 2z^2$, $N = 1 + 2z^2$ and $D = LN$.

Theorem 24 (three-row generating function, parts (b) and (c) computer-assisted)**.**

(a) The series $\sum_{m\geq 3} a_3(m)z^m$ equals $2G_{AC} + 4G_{AB} + G_X - z - 12z^2$, with G_{AB} and G_X from Propositions 28 and 27, of denominators dividing D^3 and D^2.

(b) Its reduced denominator is exactly $Q = R^3L^3N^3$.

(c) With

$$\begin{aligned} P(z) = z &- 3z^2 - 34z^3 + 228z^4 - 641z^5 + 1307z^6 - 2186z^7 \\ &+ 2056z^8 - 236z^9 - 1060z^{10} + 2584z^{11} - 4264z^{12} \\ &- 1440z^{13} + 704z^{14} - 1024z^{15} + 2080z^{16} + 2176z^{17} + 384z^{18}, \end{aligned}$$

we have

$$\sum_{m\geq 3} a_3(m)z^m = \frac{P(z)}{Q(z)} - z - 12z^2. \tag{6}$$

Proof. (a) Use (5), with $z + 12z^2$ removing widths one and two. (b) This is Theorem 29. (c) Exact computer algebra in SymPy [21] gives P, and an independent transfer matrix confirms (6) exactly through width 100 (`three_row_closed_form_check.py`). □

Let $\rho = 1 + \sqrt{3}$ and $\sigma = 1 - \sqrt{3}$.

Theorem 25 (three-row closed form, computer-assisted exact algebra)**.** *For every $m \geq 3$,*

$$E_m = p_+(m)\rho^m + p_-(m)\sigma^m + T_m, \tag{7}$$

where $432\,p_\pm(m) = (19 \pm 11\sqrt{3})m^2 + (61 \pm 41\sqrt{3})m + 104 \pm 28\sqrt{3}$ and $T_m = (-2)^{\lfloor m/2 \rfloor}\,\kappa(m)$ with $\kappa(m) = (2m^2 + 41m - 104)/216$ for even m and $\kappa(m) = (7m^2 - 32m - 175)/432$ for odd m. Hence $E_m \sim \frac{19+11\sqrt{3}}{432} m^2\rho^m$. Without the computer, we have $E_m > 0$ and

$$a_3(m) = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m + O(m^2\rho^m). \tag{8}$$

Proof. By (5), the number E_m is the coefficient of z^m in $4G_{AB} + G_X$, whose denominator divides L^3N^3. Its poles $1/\rho$, $1/\sigma$ and $\pm 1/(i\sqrt{2})$ are at most triple, so beyond a polynomial part of degree one, the sequence has the form (7). Exact computer algebra gives the twelve rational coefficients. Both sides satisfy the order-12 recurrence of L^3N^3 from $m = 14$ on and agree for $m = 2, \ldots, 13$, hence always. As $|\sigma| < 1$ and $|T_m| = O(m^2 2^{m/2})$, the asymptotic of E_m follows. Without the computer, the pole $1/\rho$ of least modulus gives $E_m = O(m^2\rho^m)$, hence (8) by (4), and the cross of Lemma 16, with word $Xa \cdots a$, gives $E_m > 0$. □

5.2 The classes AB and X

This subsection proves Propositions 27 and 28 and Theorem 29, which the proof of Theorem 24 uses. Here $D = LN = 1 - 2z - 4z^3 - 4z^4$. A *restricted run* is a string of singletons with no two letters b at distance two. Reversal and reflection carry the next lemma to BA, BC and CB.

Lemma 26 (forbidden windows in the classes containing B)**.** *In the class AB and the triple class, a window fails exactly when the table lists it for its class, with s any singleton.*

class	*windows*	*forbidden*
AB, X	*singletons only*	*bsb*
AB	*one double*	*bsA, Asb, bAb, bAc, cAb, aBa, aBb, bBa, cBc*
	two doubles	*bAB, ABa*
X	*one triple*	*bsX, Xsb, aXa, aXb, bXa, bXc, cXb, cXc*

Proof. Singleton windows follow from Lemma 20, and sAt as in Lemma 21(a), except that bAb now fails test (e) of Lemma 19 as $B \in F$. In sBt and sXt both corners in column $i+1$ are black, so the diamond is empty for $s = t = b$, and for outer s and t test (c) fails it exactly when $s = t$. For $s = b \neq t$ it shows $(i + 2, 2)$ alone, and test (e) fails it exactly when $\{2, t\} \in F$. In uAB test (d) fails $u = b$, as $(i + 2, 3)$ and $(i + 1, 2)$ are black, and otherwise three corners show. In ABw the diamond is empty for $w = b$, and test (e) fails it exactly for $w = a$. In Asb and Xsb tests (b) and (e) fail the vertical pair or the point of column $i + 1$. Other windows with a double or X at the left tip show three corners or an oblique pair needing the white point $(i + 1, 2)$. The rest is symmetric. □

Give an outer letter bit 0 and weight 2, and give b bit 1. Let h_{ij} count, by length, the continuations of a restricted run ending in bits i, j, where b cannot follow $1, j$. Read runs $w_1 \cdots w_n$ with $n \geq 1$ outward from a double. Let g_0 and g_1 count those with w_1 a prescribed outer letter and with $w_1 = b$, and u and v the same runs with w_2, if present, outer. A fictitious letter before w_1, outer for g and b for u and v, gives $(g_0, g_1, u, v) = z(h_{00}, h_{01}, h_{10}, h_{11})$. Solving the linear system for h gives

$$Dg_0 = z(1+z+3z^2+2z^3),\ \ Dg_1 = z(1+z),\ \ Du = z(1+2z^2+2z^3),\ \ Dv = z. \tag{9}$$

Proposition 27 (the triple class)**.** *With u and v as in (9), $G_X(z) = z\big(1 + 4u + 2v + 2u^2 + v^2\big)$, and its denominator divides D^2.*

Proof. In UXW the runs are restricted, and bsX and Xsb make the second letters from X outer. One nonempty run gives $2u + v$ on either side. With two, the window at X allows only the neighbor pairs (a, c), (b, b) and (c, a), which give $2u^2 + v^2$. □

A word of AB is $UAVBW$ with restricted runs U, V and W. By Lemma 26, a nonempty U read from A is counted by u or v, and a nonempty W read from B by g_0 or g_1. Let ℓ_t and r_t count the allowed U and W, empty included, when t follows A or precedes B. The windows at the double give

$$(\ell_a, \ell_b, \ell_c) = (1+2u+v, 1+u, 1+2u),\ (r_a, r_b, r_c) = (1+g_0, 1+g_0+g_1, 1+g_0+g_1). \tag{10}$$

For $s \in \{00, 10\}$ let $J_{s\beta}$ count the continuations from s ending in bit β, the empty one with $\beta = 0$. Solving as for h gives, with rows s and columns β,

$$J = \frac{1}{D}\begin{pmatrix} 1 + 2z^2 + 2z^3 & z + z^2 \\ 1 & 2z^2 + 2z^3 \end{pmatrix}. \tag{11}$$

Proposition 28 (the class $\{A, B\}$, polynomial form computer-assisted)**.** *With (9) to (11),*

$$\begin{aligned} G_{AB}(z) = {} & z^2(1 + 2u)(1 + g_0 + g_1) + z^3(\ell_a r_a + \ell_b r_b + \ell_c r_c) \\ & + z^4\,(\ell_a + \ell_c,\ \ell_b)\,J\,(r_a + r_c,\ 2r_b)^T. \end{aligned} \tag{12}$$

Moreover $G_{AB} = P_B/D^3$ with

$$P_B(z) = z^2(1 + z)(1 + z^2 - 3z^3 + 4z^6 + 4z^7). \tag{13}$$

Proof. Split by the length k of V. For $k = 0$ the windows uAB and ABw give the first term, as $u \neq b$ and $w \neq a$. For $k = 1$ the window AtB passes, and uAt and tBw give $\ell_t r_t$. For $k \geq 2$ the window Av_1v_2 makes v_2 outer, $v_{k-1}v_kB$ passes, and $v_3 \cdots v_k$ is a continuation from bits 00 or 10, as v_1 is outer or b, with weight $\ell_{v_1} r_{v_k}$. Since $r_a + r_c$ counts both outer v_k, for $k \geq 3$ the weight 2 of an outer v_k in J, like the 2 in $2r_b$, counts the choices of v_2, and $k = 2$ is the constant term of J. Exact computer algebra gives (13). □

Theorem 29 (the exact denominator, computer-assisted)**.** *The reduced denominator of* $\sum_{m\geq 3} a_3(m)z^m$ *is* $Q = R^3L^3N^3$.

Proof. By Theorem 24(a) the denominator divides $R^3D^3 = Q$. No two of R, L and N share a root, as $D(\frac{1}{3}) \neq 0$ and $L - N = -2z - 4z^2$ vanishes only where $N \neq 0$. At $z = \frac{1}{3}$ only G_{AC} has a pole, and the numerator in (2) is $\frac{4}{81}$. As $P_B \bmod L = \frac{z}{2}$ and $P_B \bmod N = \frac{1}{4} - \frac{z}{2}$, at each root of D the term G_{AB} has a triple pole, G_X at most a double pole, and G_{AC} none. □

Remark 30 (optima up to symmetry). Let $o_3(m)$ count optima up to the reflection in the middle row and the column reversal. It is 15, 58, 227, 864, 3190 and 11502 at widths 3 to 8. By Burnside's lemma, $4o_3(m)$ is $a_3(m)$ plus the optima fixed by the reflection, the reversal and the half turn. An optimum fixed by the reflection uses only b, B and X, so it is one X among letters b, and for $m \geq 4$ a window Xbb, bbX or bbb fails (Lemmas 19(b) and 20), so only bXb is fixed.

Write $m = 2k$ or $m = 2k + 1$. The half turn reverses the word and swaps A with C and a with c. A fixed optimum with two doubles thus has A and C at mirror positions $j \leq k$ and $m+1-j$, and center b for odd m. Its first k letters determine it, and only its windows centered at A or C can fail (proof of Proposition 22), those at C mirroring those at A. By Lemma 21(a), this leaves 3^{k-1} words in AC for $j = 1$, 3^{k-2} for $j = k \geq 2$, $2 \cdot 3^{k-2}$ for $1 < j < k$, and as many in CA. So for $k \geq 2$ there are $4k3^{k-2}$ of them, and for $m = 3$ there are AbC and CbA, with series $W = 2z^3 + \frac{4}{9}(1+z)\big(3z^2(1-3z^2)^{-2} - 3z^2\big)$.

A palindrome repeats each letter off its center, so the reversal fixes only words UXU^{rev} of odd width. There tXt passes only for $t = b$, and bsX and Xsb make the letter at distance two from X outer (Lemma 26), so U read outward is counted by v in (9). In the words $UX\iota(U^{\mathrm{rev}})$ fixed by the half turn, with ι the reflection, the central pairs (a, c), (b, b) and (c, a) pass, so U

is counted by $2u + v$. Adding z^3 for bXb, $2z\big(u(z^2) + v(z^2)\big)$ and W gives

$$4\sum_{m\geq 3} o_3(m)z^m = \sum_{m\geq 3} a_3(m)z^m + z^3 + \frac{4z^3(1+z^4+z^6)}{M(1+2z^4)} + \frac{2z^3(1-z)(1+5z+3z^2-3z^3)}{(1-3z^2)^2},$$

with $M = 1 - 2z^2 - 2z^4$, since $D(z^2) = M(1 + 2z^4)$. A computer check (`three_row_orbits.py`) confirms the fixed counts through width 24 and $o_3(m)$ for $m \leq 10$.

6 Four rows

Theorem 31 (four rows)**.** *For every $m \geq 6$ we have $f(4, m) = m + 4$.*

Theorem 8 gives the upper bound. For the lower bound take

$$S = \{(i, 1) : 2 \leq i \leq m - 1\} \cup \{(m - 1, 2), (m, 2), (1, 3), (2, 3), (1, 4), (m, 4)\}. \tag{14}$$

Here $|S| = m+4$ and G_S is the eight-cycle $r_1\, c_2\, r_3\, c_1\, r_4\, c_m\, r_2\, c_{m-1}$ with leaves $c_3, \ldots, c_{m-2}$ on r_1.

Proof of Theorem 31. As G_S has no four- or six-cycle, axis-aligned rectangles have distinct presentations and none is hidden, by Lemma 7.

By Lemma 15, tilted corners are black in row 1 and white in row 4. The other black corners are $(i+2, 2)$ if $i \geq m-3$ and $(i+1, 3)$ if $i = 1$ in $D(i, 1)$, $(i, 3)$ if $i \leq 2$ and $(i+1, 2)$ if $i = m - 2 \geq 4$ in $D(i, 2)$, and $(i, 3)$ if $i \leq 2$ and $(i + 3, 2)$ if $i \geq m - 4$ in A_i and B_i. By Lemma 4 a rectangle with at most one black corner is safe, leaving those below, where $s = 1$ for A_i and $s = 2$ for B_i.

rectangle	which i	corners shown
$D(1, 1)$		$(1, 2), (3, 2)$
$D(i, 1)$	$i \in \{m - 3, m - 2\}$	$(i, 2), (i + 1, 3)$
A_i or B_i	$i \in \{1, 2\}$, $i \notin \{m - 4, m - 3\}$	$(i + s, 4), (i + 3, 2)$
A_i or B_i	$i \in \{m - 4, m - 3\}$, $i \notin \{1, 2\}$	$(i, 3), (i + s, 4)$
A_i or B_i	$i = 2$, $m = 6$	$(i + s, 4)$

Shown rows differ between the first four lines, and within a line the point in row 2 or 3 fixes i and that in row 4 fixes s.

No row r has $(1, r)$ and $(3, r)$ black, so $\{(1, 2), (3, 2)\}$ has no rival. In the next three lines the axis-aligned rectangle through the shown points has white corner $(i, 3)$, $(i + s, 2)$, or $(i, 4)$, respectively, since $i \geq 3$, $i + s \leq 4 < m - 1$, and $3 \leq i \leq m - 3$ there. In the last line $m = 6$, and the shown point $(z, 4)$ has $z \in \{3, 4\}$ and no rival, as column z is black only in row 1 and no column is black in rows 1 and 4. □

By Theorem 11, for $m \geq 4$ the axis-valid blackouts with $m + 4$ points are the $3\,(m)_4\,4^{m-4}$ *candidates*, with four doubles whose row pairs form a four-cycle F and singletons elsewhere.

Corollary 32 (widths four and five, computer-assisted)**.** *We have* $f(4, 4) = 7$ *and* $f(4, 5) = 8$*.*

Proof. Lemma 16 gives the lower bounds, and a computer finds a repeated presentation in all 72 and 1,440 candidates. □

Theorem 33 (no hidden rectangle)**.** *For* $m \geq 4$*, no valid blackout of the* $4 \times m$ *grid hides a rectangle.*

Proof. By Lemma 2 it suffices to write each corner set as $C(R_1) \triangle C(R_2)$ with $R_1 \neq R_2$. For an axis-aligned rectangle, keep its rows and pair each of its columns x, y with a third column w. For $1 \leq i \leq m - 3$ let $\mathsf{Q}_i = \{(i, 2), (i, 3), (i + 3, 2), (i + 3, 3)\}$. Then

$$\begin{aligned} \mathsf{C}_i = \mathsf{A}_i \triangle \mathsf{Q}_i, \qquad & \mathsf{E}_i = \mathsf{B}_i \triangle \mathsf{Q}_i, \\ D(i + 1, 1) = D(i, 2) \triangle \mathsf{A}_i, \qquad & D(i + 1, 2) = D(i, 1) \triangle \mathsf{E}_i, \end{aligned} \tag{15}$$

and each solves for any of its three rectangles. The first line covers A_i, B_i, C_i, and E_i, and the second, with $i = j - 1$ or $i = j$, covers $D(j, 1)$ and $D(j, 2)$, since $j \geq 2$ or $j \leq m - 3$ as $m \geq 4$. □

Theorem 34 (forbidden-block bound)**.** *For every* $m \geq 6$ *we have* $a_4(m) \leq 3\,(m)_4\,4^{m-4}(53/64)^k$*, where* $k = \max(0, \lfloor m/4 \rfloor - 4)$*. So* $a_4(m) = O\big(m^4\,212^{m/4}\big)$*.*

Proof. Optima are candidates, with $3\,(m)_4$ choices of F, the double columns, and their pairs. A *block* is a run of four singleton columns, whose word lists their black rows. For a block from column x, the *template words* 2413, 3412,

2143, and 3142 are the corner rows of C_x, A_x, E_x, and B_x. Let T be one, with word $t_1t_2t_3t_4$.

A *swap word* swaps two letters t_p and t_q. No block carries one, since T and the axis-aligned rectangle on $(x+p-1, t_p)$ and $(x+q-1, t_q)$ would both show exactly these two points. Template words are the permutations with both end letters in $\{2, 3\}$. Swapping two end or two middle letters gives one again, and swapping an end and a middle letter gives the sixteen with one end letter in $\{2, 3\}$, so the 20 swap words are these and the four template words.

An *edit word* replaces a letter $t_p = r$ by s with $\{r, s\} \in F$. No block carries one, since T and the rectangle on rows r, s and columns $x+p-1$ and d, the double column of $\{r, s\}$, would both show only $(x + p - 1, r)$. Each template word has 8 edit words. Two template words share one only if they differ in two positions holding r, s in opposite orders, and then share two if $\{r, s\} \in F$, which happens for two pairs with $\{r, s\} = \{2, 3\}$ and two with $\{1, 4\}$. So there are at least $32 - 8 = 24$ edit words. They repeat a letter, so none is a swap word, and at most $256 - 20 - 24 = 212 = 4^4 \cdot 53/64$ words are allowed in a block.

At most four of the $\lfloor m/4 \rfloor$ column groups $4j-3, \ldots, 4j$ contain a double column, so at least k are blocks. Other columns have at most 4 values. □

Lemma 35 (six columns of memory)**.** *For $m \geq 6$ and a pair set F, a word over the four singleton letters and the pairs of F, each pair used once, is the word of an optimum exactly when (i) no tilted rectangle is hidden, (ii) none shares its presentation with an axis-aligned rectangle whose columns are among its own and the double columns, and (iii) no two tilted rectangles within seven consecutive columns share a presentation.*

Proof. Optima satisfy the conditions, (i) by Theorem 33. Conversely, the word is axis-valid by Theorem 11, and a hidden axis-aligned rectangle would be a four-cycle of G_S. Given (i), two distinct rectangles with one presentation P have $P \neq \emptyset$, and one is tilted. If the other is axis-aligned, each of its columns meets P, so it is a column of the tilted one or, with two black corners, a double column, against (ii). If both are tilted, they share a column c meeting P. Each spans at most four consecutive columns, so both lie in columns $c - 3$ to $c + 3$, against (iii). □

Theorem 36 (the four-row generating function, computer-assisted)**.** *There are coprime integer polynomials $P_4 = 26z^6 - 826z^7 + \cdots$ of degree 444 and*

$Q_4 = \sum_{j=0}^{420} q_j z^j$ *of degree* 420 *with* $q_0 = 1$ *and* $\sum_{m \ge 6} a_4(m) z^m = P_4/Q_4$. *The recurrence* $\sum_j q_j\, a_4(n-j) = 0$ *holds for every* $n \ge 445$ *and fails at* $n = 444$. *Over the integers*

$$Q_4 = (1-z)^2 (1+z)^2\, g_5^3\, \tilde{g}_5^3\, g_6^3\, g_{11}^5\, g_{17}^3\, g_{19}^3\, g_{20}^5\, g_{21}^5,$$

where g_d *and* $\tilde{g}_d$ *are irreducible of degree* d *with constant term* 1.

Proof. By Lemmas 35 and 17, an automaton remembering the last six columns and the used pairs of F accepts exactly the optima with pair set F. Two transfer counters built from the definition [26], one checked against this automaton, agree on $a_4(m)$ for $m \le 1500$ and with Table 1. Berlekamp–Massey modulo eight primes gives P_4 and Q_4. The counter checked against the automaton is a column machine with a register of used pairs. It has a transfer matrix T_F for each of the three pair sets F, the four-cycles with parts $\{1,2\}$ and $\{3,4\}$, $\{1,3\}$ and $\{2,4\}$, or $\{1,4\}$ and $\{2,3\}$, with 2,519, 3,089, and 2,119 reachable states. Its state vectors $v(n) = v(0) T_F^n$ satisfy $\sum_j q_j v(n_F - j) = 0$ at $n_F = 436$, 435, and 445, so powers of T_F give the recurrence for $n \ge 445$, and $\deg P_4 = 444$ forces its failure at $n = 444$. SymPy factors Q_4. □

Corollary 37 (growth, computer-assisted)**.** *Let* $\rho_4 = 3.351996777310\ldots$ *be the largest real root of* $p(x) = x^{20} g_{20}(1/x) = x^{20} - 4x^{19} + 2x^{18} + \cdots - 26x^2 - 4x + 8$. *There is a polynomial* B *of degree* 4 *with* $a_4(m) = \rho_4^m B(m) + O((1000/341)^m)$, *so* $a_4(m) \sim c\, m^4 \rho_4^m$ *with* $c \approx 8.1611 \times 10^{-5}$.

Proof. Exact root isolation on the factors of Q_4 shows that $r = 1/\rho_4 = 0.29832\ldots$ is its only root of modulus at most 0.341, simple in g_{20} and so of order five. So $P_4/Q_4 = \sum_{j=1}^{5} c_j (1 - z/r)^{-j} + E(z)$, where E has coefficients $O((1000/341)^m)$ and, by coprimality, $c_5 = H(r) \ne 0$ for $H(z) = P_4(z)(1 - z/r)^5/Q_4(z)$. As $[z^m](1 - z/r)^{-j} = \rho_4^m \binom{m+j-1}{j-1}$, we get $B(m) = \sum_j c_j \binom{m+j-1}{j-1}$ and $c = c_5/24$. Interval arithmetic puts c between $8.1610970775 \times 10^{-5}$ and $8.1610970776 \times 10^{-5}$. □

7 Five and six rows

7.1 Boundary padding

For a blackout S of the $h \times m$ grid and $L, R \ge 0$, let $S_{L,R}$ add the columns $1 - L, \ldots, 0$ and $m + 1, \ldots, m + R$, each black exactly in the *padding row*, row 1 or row h.

Theorem 38 (finite collar)**.** *Let $h \geq 3$, fix the padding row and the padded sides, and let L^* and R^* be W_h of Lemma 14 on padded sides and 0 otherwise. Then $S_{L,R}$ is valid for all $L, R \geq 0$ vanishing on unpadded sides if and only if S_{L^*,R^*} is valid.*

Proof. Write $S^* = S_{L^*,R^*}$. One direction is clear. Let S^* be valid and $U = S_{L,R}$. By Lemma 13 and (1), a tilted rectangle has at most one corner in row 1 and at most one in row h. Hence a weak tilted rectangle of U has a black corner in an original column e, and by Lemma 14 its columns lie in $[e - W_h, e + W_h] \cap [1 - L, m + R] \subseteq [1 - L^*, m + R^*]$. As U and S^* agree on their common columns, it is a rectangle of S^* with the same presentation. Suppose distinct rectangles R_1 and R_2 of U have the same presentation P.

Step 1. Both axis-aligned. Added columns are vertices of degree one, so the cycles of G_U lie in G_{S^*}. As S^* is valid with at least three columns, U is axis-valid by Lemma 7.

Step 2. Both tilted. By Lemma 4, $|P| \leq 2$, so both are weak, hence rectangles of S^* with the same presentation, a contradiction.

Step 3. One of each. Let R_1 be tilted, hence a rectangle of S^*. A column of R_2 that meets P is a column of R_1, and one that does not has two black corners and is an original column. So R_2 is a rectangle of S^* too, a contradiction. □

7.2 Five rows

Let S_8 be the blackout of the 5×8 grid with $B_1 = \{1, 4\}$, $B_2 = \{4, 5\}$, $B_3 = \{2, 5\}$, $B_4 = B_5 = \{5\}$, $B_6 = \{3, 4\}$, $B_7 = \{2, 3\}$, and $B_8 = \{1, 2\}$.

Theorem 39 (five rows, (a) and (c) computer-assisted, (b) checked in Lean)**.**

(a) We have $f(5, m) = m + 5$ for $m = 5$, 6, and 7.

(b) We have $f(5, m) = m + 6$ for every $m \geq 8$, attained by S_8 followed by $m - 8$ columns black in row 1 only.

(c) The 5×8 grid has exactly four maximum blackouts, the images of S_8 under row reflection and column reversal.

Proof. (b) The upper bound is Theorem 8. Since $W_5 = 5$, Theorem 38 with padding row 1 on the right reduces S_8 to the test board of Figure 2(a). Its 954 rectangles have distinct presentations, by three enumerators and Lean.

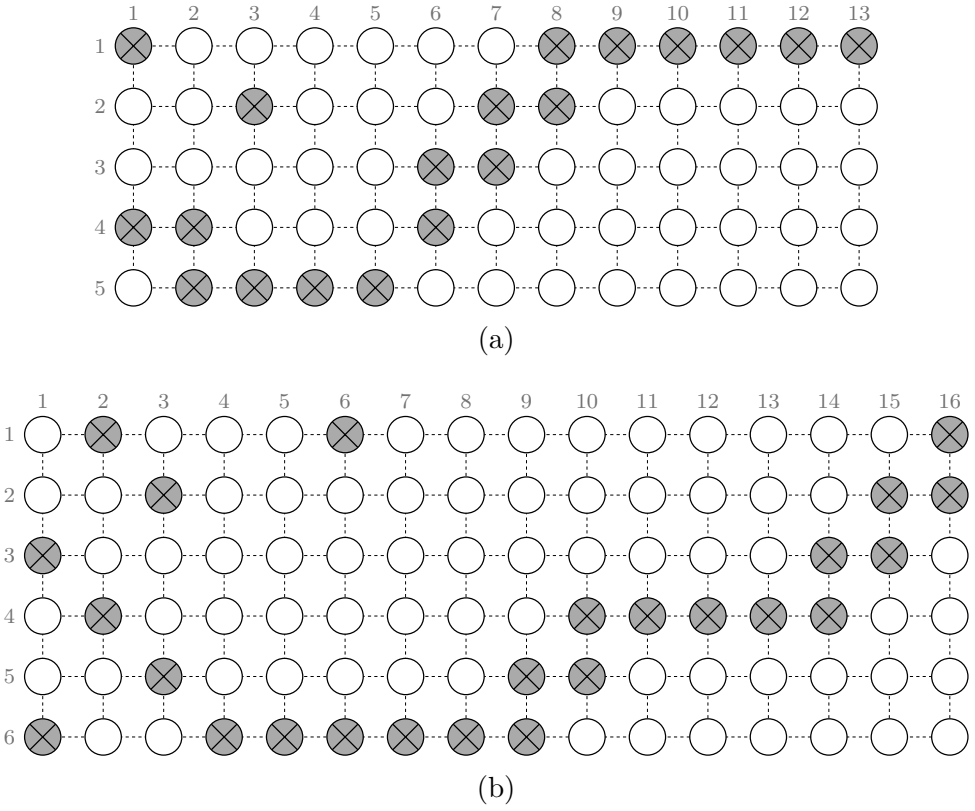


Figure 2: (a) The 5×13 test board of S_8, whose last five columns are black only in row 1. (b) The 6×16 blackout S_{16}. Its test board adds seven columns black only in row 1 on the right.

(a) For $m = 5$ the bounds are Corollary 12 and S_0 of Theorem 43. For $m = 6, 7$ the stored blackouts in `jis/data/small_cells_sat_5x6.json` and `jis/data/small_cells_sat_5x7_b12.json` give the lower bound. Checked SAT certificates show that no blackout of $m + 6$ points is valid (Section 9).

(c) Two SAT enumerations and a column search over the $10\,(8)_6\,5^2 = 5{,}040{,}000$ candidates of Theorem 11 find exactly the four images. □

By Theorem A, $f(h, m) \le h + m$ when a side is at most four, and also on the 5×5, 5×6, 5×7, and 6×6 grids. These are the other grids of area below 40, so the guess $f(h, m) = h + m$ first fails at 5×8.

Remark 40 (five-row counts, computer-assisted). For $m = 13, \ldots, 23$ the

counts $a_5(m)$ are

$$193{,}406,\quad 1{,}346{,}758,\quad 8{,}650{,}416,\quad 52{,}128{,}242,$$
$$299{,}349{,}210,\quad 1{,}653{,}637{,}492,\quad 8{,}828{,}301{,}226,\quad 45{,}743{,}165{,}894,$$
$$230{,}934{,}909{,}328,\quad 1{,}139{,}486{,}126{,}866,\quad 5{,}510{,}071{,}794{,}736.$$

Two independent transfer counters based on Theorem 11 agree on each.

7.3 Six rows and small grids

Let S_{16} be the blackout of the 6×16 grid in Figure 2(b). Its doubles, in the columns $1, 2, 3, 6, 9, 10, 14, 15, 16$, carry the edges of $K_{3,3}$ between $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

Theorem 41 (six rows, (a) computer-assisted, (b) checked in Lean)**.**

(a) For $8 \le m \le 14$ we have $f(6, m) \le m + 8$.

(b) For every $m \ge 15$ we have $f(6, m) = m + 9$.

Proof. (a) For $m = 8$ this is Corollary 12. For $9 \le m \le 14$, by Theorem 11 the doubles of a valid blackout of $m+9$ points carry the edges of $K_{3,3}$ for one of ten row partitions. Checked LRAT certificates from two SAT encoders exclude every case.

(b) The upper bound is Theorem 8. For $m = 15$ the blackout with $B_1 = \{5, 6\}$, $B_2 = \{4, 5\}$, $B_3 = \{3, 4\}$, $B_4 = \{2, 3\}$, $B_5 = B_6 = \{1\}$, $B_7 = \{1, 4\}$, $B_8 = B_9 = \{1\}$, $B_{10} = \{1, 2\}$, $B_{11} = \{3, 6\}$, $B_{12} = \{1\}$, $B_{13} = \{2, 5\}$, $B_{14} = \{6\}$, and $B_{15} = \{1, 6\}$, stored in `jis/data/six_row_sat_w15_A.json`, gives the lower bound with 24 points. For $m \ge 16$, since $W_6 = 7$, the test board of Theorem 38 adds $(17, 1), \ldots, (23, 1)$ to S_{16} of Figure 2(b). Its 4,627 rectangles have distinct presentations, by three enumerators and Lean. □

Theorem 42 (small grids, computer-assisted)**.** *We have $f(6, 6) = 12$, $f(6, 7) = 13$, $f(7, 7) = 14$, and $f(6, 8) = 15$. The blackout $(1, 1)$, $(1, 5)$, $(2, 1)$, $(3, 1)$, $(3, 2)$, $(3, 3)$, $(4, 3)$, $(4, 4)$, $(5, 4)$, $(6, 4)$, $(7, 4)$, $(7, 5)$, $(7, 6)$, $(8, 2)$, $(8, 6)$ of the 6×8 grid attains the last value.*

Proof. The stored blackouts in `jis/data/small_cells_sat_6x6.json`, `jis/data/small_cells_sat_6x7.json`, and `jis/data/small_cells_sat_7x7.json` give the other lower bounds. Checked SAT certificates exclude one more point in each case. □

Theorem 43 (the 5×5 grid)**.**

(a) We have $f(5,5) = 10$.

(b) (Computer-assisted.) The $a_5(5) = 8$ maximum blackouts are the images of S_0 below under the square's symmetries.

Proof. (a) The upper bound is Corollary 12. For the lower bound let S_0 have $B_1 = \{4,5\}$, $B_2 = \{3,4\}$, $B_3 = \{1,2,3\}$, $B_4 = \{1\}$, and $B_5 = \{1,5\}$. Its graph is an 8-cycle with two pendant edges, so S_0 is axis-valid. By Lemma 13 its 30 tilted rectangles are nine unit diamonds, sixteen from Lemma 15, and five spanning the grid. The fifteen weak ones are listed by their two shown corners and, after a bar for an oblique pair, the white corner of its axis-aligned rectangle.

$(1,2),(2,1) \mid (1,1)$	$(1,3),(2,2) \mid (1,2)$	$(3,4),(4,3) \mid (4,4)$	$(2,5),(3,4) \mid (3,5)$
$(2,2),(4,2)$	$(1,2),(4,3) \mid (1,3)$	$(4,4),(5,2) \mid (4,2)$	$(3,5),(4,4) \mid (3,4)$
$(4,3),(5,2) \mid (4,2)$	$(3,4),(5,2) \mid (5,4)$	$(2,5),(5,2) \mid (2,2)$	$(4,5),(5,3) \mid (4,3)$
$(1,3),(4,2) \mid (1,2)$	$(2,5),(4,3) \mid (4,5)$		$(4,5),(5,4) \mid (4,4)$

The fifteen pairs are distinct, and Lemma 4 settles the other tilted rectangles. Each listed white corner lies on the only axis-aligned rectangle through its pair, and $(2,2),(4,2)$ has no rival since $B_2 \cap B_4 = \emptyset$. So S_0 is valid.

(b) Of the 44,640 sets of 10 points whose graph G_S has no four- or six-cycle, exactly the images of S_0 are valid, by three programs. □

8 Every fixed height

Fix h. A blackout of the $h \times m$ grid is a *column word* $w = B_1 B_2 \cdots B_m$ over the subsets of $\{1, \ldots, h\}$, with length $|w|$ and blackout $S(w)$. Let $\mathcal{L}_h$ be the set of words with a valid blackout, the empty word included.

Theorem 44 (regularity and rational series)**.** *For every $h \geq 1$ the language $\mathcal{L}_h$ is regular, and $G_h(z,u) = \sum_{w \in \mathcal{L}_h} z^{|w|} u^{|S(w)|}$ is rational in z and u.*

Proof. The *trace* of a rectangle R is the word whose ith letter is the set of rows of its corners in column i, and it determines $C(R)$. By Lemma 2, $w \notin \mathcal{L}_h$ exactly when two different traces Y, Y' of length $|w|$ have $Y_i \triangle Y'_i \subseteq B_i$ for every i. The traces form a regular language, $\emptyset^*\{r,s\}\emptyset^*\{r,s\}\emptyset^*$ for axis-aligned rectangles and, by Lemmas 13 and 14, $\emptyset^* Y_0 Y_1 \cdots Y_n \emptyset^*$ for finitely many templates (1) with $n \leq W_h$. So an automaton that reads w and two

traces in parallel recognizes the complement of $\mathcal{L}_h$. Weighting a letter B by $u^{|B|}$, the transfer matrix $M(u)$ of a deterministic automaton for $\mathcal{L}_h$ gives $G_h = \alpha(I - zM(u))^{-1}\beta$, rational by Cramer's rule [9, Proposition V.6]. □

Theorem 45 (fixed height)**.** *Let $h \geq 2$. (a) There are integers $m_0 \geq 3$, $p \geq 1$, and $k_0, \ldots, k_{p-1}$ such that $f(h, m) = m + k_{m \bmod p}$ for every $m \geq m_0$. (b) The series $\sum_{m \geq 0} a_h(m) z^m$ is rational, with the conventions of Section 2, and the words of optimum blackouts form a regular language.*

Proof. Let $E(w) = |S(w)| - |w|$. By Theorem 8 and $|S(v)| \leq h|v|$, every factor $v = B_i \cdots B_j$ of a valid word has $E(v) \leq C_h = \max(q_h, 2h - 2)$. So a valid $w = uv$ of excess k has $k - C_h \leq E(u) \leq C_h$. An automaton for $\mathcal{L}_h$ with a counter for the excess of prefixes, kept in this range, accepts the set $\mathcal{L}_{h,k}$ of valid words of excess k, so the lengths of these words are eventually periodic. For $m \geq 3$, Lemma 16 and Theorem 8 make $f(h, m) - m$ the largest k from $h - 1$ to q_h with a word of length m in $\mathcal{L}_{h,k}$, which gives (a). The optimum words are those of $\mathcal{L}_{h,k_r}$ of lengths $m \geq m_0$ with $m \equiv r \pmod{p}$, and finitely many shorter ones, so (b) follows as in Theorem 44. □

9 Computations and verification

Each computed value and count comes from two separately written programs, except the 7×8 count below. Code, data, and commands are in the repository [26], and the paths below are relative to its folder `kagey-problems/problem001`. For each computer-assisted result, `jis/scripts/README.md` gives the command, the evidence file with its SHA-256 hash, and the stored time. A longer version of this paper in the repository [27] has fuller proofs of the four-row computations and of the verification, and `jis/removed` keeps text cut from earlier drafts.

An ILP program with CBC [10] through PuLP [22] finds a smallest hitting set for the sets of Lemma 2, and brute force checks it. The SAT solvers of the searches and counts come from PySAT [14]. SAT encoding A forbids each minimal symmetric difference of two rectangles. Encoding B has a clause for each pair of rectangles with a common corner and an indicator for each hidden rectangle, at most one of them true. The structured encodings for six rows add the column structure of Theorem 11 for one row partition. Column searches add columns from left to right, using $f(6, 6) \leq 12$ on the 6×7 and 6×8 grids and $f(6, 7) \leq 13$ on the 6×8 and 7×7 grids.

grid	f	$a_h(m)$	by
2×2	4	1	HIB
2×3	4	12	HIB
2×4	5	32	HIB
2×5	6	80	IB
2×6	7	192	TI
2×7	8	448	TI
3×3	5	53	IB
3×4	6	224	IB
3×5	7	892	IB
3×6	8	3,420	IB
3×7	9	12,704	IB

grid	f	$a_h(m)$	by
3×8	10	45,864	IB
3×9	11	161,992	IB
3×10	12	562,912	ITD
4×4	7	316	IB
4×5	8	1,810	IB
4×6	10	26	IB
4×7	11	292	IB
4×8	12	2,118	IMD
5×5	10	8	IB
5×6	11	76	ISC
5×7	12	1,184	ISC

grid	f	$a_h(m)$	by
5×8	14	4	TSDC
5×9	15	44	TSC
5×10	16	350	TSC
5×11	17	3,004	TSC
5×12	18	24,706	TSCM
6×6	12	160	ISC
6×7	13	2,536	SC
6×8	15	8	SC
6×15	24	–	TS
7×7	14	7,300	SC
7×8	16	–	S

Table 1: Computed values for $h \leq m$. H: hand, T: theorem, I: ILP (value only on 3×10, 4×8, 5×6, 5×7, and 6×6), B: brute force, D: equality candidates, C: column search, S: two SAT encodings, listing all optima where a count is given, M: transfer counter, –: not established.

Our rectangle lists agree with A085582 on squares and A289832 on the other grids.

CaDiCaL [4] gives LRAT certificates for $f(5,6) < 12$, $f(5,7) < 13$, $f(6,6) < 13$, $f(6,7) < 14$, $f(7,7) < 15$, $f(6,8) < 16$, $f(7,8) < 17$, and the 70 six-row instances (widths 8 to 14, ten partitions each) behind Theorem 41(a). The seven grids use encoding A over all blackouts, and the six-row instances use the structured encoding B. We split the 6×8 instance into the 64 cases of its first column, and the 7×7 and 7×8 instances into 128, and the second generator checks that the cases cover every first column. Two separate generators give the same clauses, and all 394 proofs pass `lrat-check` of drat-trim [5, 30] and the Lean LRAT checker. The encodings are argued faithful in `jis/data/unsat_certificates.json`, not formally. The models in `jis/data/table_counts/sat_7x8_16.json` give $f(7,8) \geq 16$, so $f(7,8) = 16$. One enumeration finds 18 optima on this grid, and a second did not finish.

The Lean 4 files in `../lean/KageyProblems/Problem001` of the repository prove Lemmas 2, 4, and 14, Corollary 3, Theorems 5, 8 (both orientations), and 38, and that a valid blackout hides at most one rectangle. The kernel checks that S_0, S_8, the blackout of Theorem 42, the 6×15 blackout of Theorem 41, and the 5×13 and 6×23 test boards are valid, and that $\{(1,2), (2,1), (3,2), (2,3)\}$ hides $D(1,1)$ on the 4×3 grid, so Theorem 33 needs $m \geq 4$. Hence Lean proves $f(5,m) = m + 6$ for $m \geq 8$, $f(6,m) = m + 9$ for $m \geq 15$, $f(5,5) \geq 10$, and $f(6,8) \geq 15$, from the three standard Lean

axioms alone. It found no error. Before it, Lemma 4 and Theorems 8 and 38 had only hand proofs, and the test boards had only Python checks. Not formalized are Lemma 13, Theorem 11, $f(5,5) \le 10$, the faithfulness of the encodings, the rest of Sections 3 to 8, and Appendix A.

10 Open questions

(1) Find the rational series $\sum_{m\ge 0} a_5(m) z^m$ (Theorem 45, Remark 40) and the growth rate of $a_5(m)$.

(2) A valid 6×9 blackout of 16 points passes (computer-checked, `six_row_collar_6x9_check.py`) the test of Theorem 38 with seven columns black only in row 1 on each side. With Theorem 41 this gives $m+7 \le f(6,m) \le m+8$ for $9 \le m \le 14$. Which value holds?

(3) Is $f(h,m) = m + q_h$ for all large m when $h \ge 7$? By Theorem 38, one valid $7 \times m$ blackout of $m+12$ points passing the collar test would settle $h = 7$.

(4) Lemma 7 and Neuwirth's bound for bipartite graphs of girth eight [24, Corollary 16] give $f(n,n) < n^{4/3} + \frac{2}{3}n - \frac{2}{9}n^{2/3} - \frac{20}{81}n^{1/3}$ for $n \ge 3$, and Theorem 48 gives $f(n,n) \ge 0.29\, n^{6/5}(1+\ln n)^{-2/5}$ for large n. Lavrov [15] used the same method for the triangular version. What is the true order, and the least n with $f(n,n) > 2n$?

11 Acknowledgments

I thank Peter Kagey for posing the problem, and for discussing its early stages with me, including the uniqueness of the 5×5 optimum and the first graph-theoretic approach to the narrow grids. The design of Figures 1 and 2 follows the figure in his posting of Problem 001 [17].

A note on the use of AI

I began this work in August and wrote the paper and its mathematics myself, together with most of the Python code. In October I used an AI tool (Claude Opus 5.5, from Anthropic) to revise the finished draft. I used it

to proofread, revise and shorten the text, and to draw the figures in the style of Kagey's posting of the problem. I used it to check my Python code, and it wrote further checks of the computer-assisted results. One of these, the script `six_row_collar_6x9_check.py`, gives the lower bound in the second open question of Section 10. It also wrote the Lean files, the checked certificates for the SAT bounds and the formal checks of Section 9. The formalization did not change any statement of the paper, and it found no error or missing hypothesis. It changed how several results are established, as Section 9 describes. Finally, I used it to search the literature beyond the sources I had found, and several references come from that search. I read every source that the paper cites, together with the parts that support each citation. I checked every change, and I take full responsibility for the contents of the paper.

A Square grids

On narrow grids $f(h, m) - m$ stays bounded. Here we show that on square grids $f(n, n)$ grows faster than any multiple of n, so the guess $f(h, m) = h+m$ fails badly there.

Let $n \geq 2$, $\Lambda = 1 + \ln n$, $\mathcal{R}$ the set of rectangles of the $n \times n$ grid, tilted ones included, and d_v the number with corner v.

Lemma 46. *If rectangles $R_1 \neq R_2$ have exactly two common corners p and q, and $\{p, q\}$ is a side of both or a diagonal of both, then $C(R_1) \triangle C(R_2)$ is the corner set of a rectangle.*

Proof. For a common side, the other corners are $p + u, q + u$ and $p + v, q + v$ with $u \neq v$ nonzero and perpendicular to $q - p$, so they are distinct and form a rectangle with sides $q - p$ and $v - u$. For a common diagonal, the other corners are $o \pm u$ and $o \pm v$ with $o = (p + q)/2$, $|u| = |v|$, and $v \neq \pm u$. The segments $[o - u, o + u]$ and $[o - v, o + v]$ share midpoint and length, so they are diagonals of a rectangle. □

Let $r_2(t)$ count the integer pairs (a, b) with $a^2 + b^2 = t$, and $\tau(t)$ the divisors of t. By Hardy and Wright [13, Sections 16.9, 16.10, and 18.1], $r_2(t) \leq 4\tau(t)$, and each $\varepsilon > 0$ has a K_ε with $\tau(t) \leq K_\varepsilon t^\varepsilon$ for $t \geq 1$.

Let $\mathcal{A}$ consist of the sets $C(R)$ with $R \in \mathcal{R}$ and the sets $C(R_1) \triangle C(R_2)$ whose exactly two common corners form a side of one and a diagonal of the other, and $\mathcal{B}$ of the sets $C(R_1) \triangle C(R_2)$ with exactly one common corner.

Lemma 47.

(a) Each point v has $d_v \le 8n^2\Lambda$, and $|\mathcal{R}| \le 2n^4\Lambda$.

(b) For every $\varepsilon > 0$, $|\mathcal{A}| \le A_\varepsilon n^{4+2\varepsilon}\Lambda$ with $A_\varepsilon = 2(1 + 16 \cdot 2^\varepsilon K_\varepsilon)$, and $|\mathcal{B}| \le 32n^6\Lambda^2$.

(c) A blackout that contains no member of $\mathcal{A} \cup \mathcal{B}$ is valid.

Proof. (a) The sides at v are $k(a,b)$ and $l(-b,a)$ with $\gcd(a,b) = 1$, $k \ge 1$, and $l \ne 0$. With $d = \max(|a|,|b|)$ there are $8d$ choices of (a,b), at most $(n-1)/d$ of k, and $2(n-1)/d$ of l. Each rectangle counts twice, so $2d_v \le \sum_{d=1}^{n-1} 16(n-1)^2/d \le 16n^2\Lambda$, and $4|\mathcal{R}| = \sum_v d_v \le 8n^4\Lambda$.

(b) A rectangle with diagonal $\{p,q\}$ is fixed by its other diagonal, an integer vector of squared length $t = |p-q|^2 < 2n^2$, so each side of each $R \in \mathcal{R}$ is a diagonal of at most $r_2(t) \le 4K_\varepsilon(2n^2)^\varepsilon$ rectangles. Hence $|\mathcal{A}| \le |\mathcal{R}|(1 + 16K_\varepsilon 2^\varepsilon n^{2\varepsilon}) \le A_\varepsilon n^{4+2\varepsilon}\Lambda$. Members of $\mathcal{B}$ come from two rectangles at a corner v, so $|\mathcal{B}| \le \sum_v \binom{d_v}{2} \le \frac{n^2}{2}(8n^2\Lambda)^2$.

(c) By Lemma 2, an invalid blackout S contains $C(R_1) \triangle C(R_2)$ for some $R_1 \ne R_2$, which share at most two corners by Lemma 4. With two, this set lies in $\mathcal{A}$ by Lemma 46 or by definition; with one, in $\mathcal{B}$; with none, $C(R_1) \subseteq S$ lies in $\mathcal{A}$. □

Theorem 48 (deletion method)**.** *We have*

$$\liminf_{n\to\infty} \frac{f(n,n)\,(1+\ln n)^{2/5}}{n^{6/5}} \ge \frac{5}{6 \cdot 192^{1/5}} = 0.2911\ldots.$$

In particular $f(n,n) - 2n \to \infty$.

Proof. Following Alon and Spencer [2, Section 3.2], put $a = 192^{-1/5}$, $T = n^{6/5}\Lambda^{-2/5}$, and $\theta = a\,n^{-4/5}\Lambda^{-2/5} \in (0,1)$, and select grid points independently with probability θ. Let X, Y, and Z count the selected points and the fully selected members of $\mathcal{A}$ and $\mathcal{B}$. Then

$$\mathbb{E}(X - Y - Z) = n^2\theta - |\mathcal{A}|\theta^4 - |\mathcal{B}|\theta^6.$$

Here $n^2\theta = aT$, and Lemma 47(b) with $\varepsilon = 1/10$ gives $|\mathcal{B}|\theta^6 \le 32a^6T$ and $|\mathcal{A}|\theta^4 \le A_{1/10}a^4n^{-1/5}T$. Since $32a^5 = \frac{1}{6}$, some selection S' has $X - Y - Z \ge (\frac{5}{6}a - A_{1/10}a^4n^{-1/5})T$. Deleting a point of each member of $\mathcal{A} \cup \mathcal{B}$ inside S' leaves at least $X - Y - Z$ points, valid by Lemma 47(c). So $f(n,n)/T \ge \frac{5}{6}a - A_{1/10}a^4n^{-1/5} \to 5/(6\cdot 192^{1/5})$, and $T/n \to \infty$ gives $f(n,n) - 2n \to \infty$. □

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